

Module 4: Two-Dimensional Ballistic-Diffusive Thermal Transport in Silicon

From microscopic carrier transport to a reduced 2D thermal model for
RadiationEffects_proj2

Physics note for RadiationEffects_proj2

Purpose. This note develops the ballistic-diffusive heat-conduction model from first principles. The goal is not only to write down the final partial differential equation, but to understand why it has the form that it does and how it should be adapted into Module 4 of RadiationEffects_proj2. We begin with conservation of energy and the microscopic carrier picture, derive the Boltzmann transport equation under the relaxation-time approximation, recover Fourier conduction as a limiting approximation, explain why Fourier and Cattaneo models fail in the quasi-ballistic regime, and then derive the ballistic-diffusive equation used here as the thermal-physics backbone. The later sections reinterpret the model for a two-dimensional silicon device cross section with volumetric source terms appropriate for radiation-damage studies and with coupling to the defect-evolution and device-transport modules.

Contents

Symbols and acronyms used in this document	5
1 What problem this module is solving	8
2 Heat carriers and the meaning of temperature	9
2.1 Heat is carried by microscopic excitations	9
2.2 Temperature is simple only near local equilibrium	9
3 Macroscopic energy conservation	10
4 Fourier conduction as a local constitutive model	10
4.1 Why the Fourier heat equation has infinite propagation speed	11
5 The Knudsen number and transport regimes	11
6 Microscopic starting point: distribution function	12

7	Deriving the Boltzmann transport equation	12
7.1	Streaming without scattering	13
7.2	Adding scattering	13
7.3	Term-by-term meaning	13
8	Why a direct Boltzmann solve is expensive	14
9	Intensity notation	14
9.1	Definition of spectral intensity	14
9.2	Moments of intensity	15
10	Recovering Fourier conduction from the BTE	15
11	Cattaneo's correction and its limitation	16
11.1	Flux relaxation	16
11.2	Why Cattaneo is still not enough	17
12	Ballistic-diffusive idea	17
13	Equation for the ballistic component	18
13.1	Derivation along a characteristic	18
13.2	Retarded boundary time	18
14	Equation for the medium/scattered component	19
15	P1 angular expansion for the medium component	19
16	Moment equations for the medium component	20
16.1	Zeroth angular moment	20
16.2	First angular moment	20
17	Heat flux and internal energy split	20
18	Constitutive relation for the medium component	21
19	Energy equation for the medium component	21

19.1	Ballistic moment relation	22
19.2	Eliminating the medium heat flux	22
20	Explicit ballistic heat flux	23
21	Boundary condition for the medium component	23
22	Temperature consistency at boundaries	24
23	One-dimensional thin-film validation problem	24
23.1	Why this validation problem is the right first test	24
23.2	Starting from the full ballistic-diffusive energy equation	25
23.3	Initial and boundary data in physical variables	26
23.4	Nondimensional variables	26
23.5	Step-by-step nondimensionalization of the governing equation	27
23.6	Dimensionless initial and boundary conditions	28
23.7	Ballistic intensity in one dimension	29
23.8	Finite arrival time and the lower angular limit	30
23.9	Ballistic flux contribution for the step-heated film	30
23.10	Ballistic internal energy contribution	31
23.11	Equilibrium subtraction and why it is not cosmetic	31
23.12	What should be plotted in the Module 4 validation	32
24	Dedicated comparison: why Fourier and Cattaneo fail	32
24.1	Failure mode of Fourier conduction	32
24.2	Failure mode of Cattaneo conduction	32
24.3	Why the ballistic-diffusive model is better	33
25	Weak form for FEM implementation	33
26	Matrix form	34
27	Module 4 FEM specialization for the MATLAB implementation	34
27.1	Strong form solved by the FEM path	35

27.2	Triangular mesh and nodal temperature unknowns	35
27.3	Weak form in temperature variables	36
27.4	Element heat-capacity matrix	36
27.5	Element conductivity matrix	37
27.6	Element source vector	37
27.7	Global semi-discrete FEM equation	37
27.8	Backward-Euler time stepping used in MATLAB	38
27.9	How the ballistic term is represented in the first FEM path	38
27.10	Verification tests for the FEM path	39
27.11	MATLAB implementation map	39
28	Practical implementation strategy for this project	39
29	How this connects to the superconducting-qubit project	40
30	Assumptions and limitations	40
31	Summary of the derivation	41
32	Lecture-ready slide translation for Module 4	42
33	Module 4 implementation checklist	43
34	Module 4 interpretation for RadiationEffects_proj2	44
34.1	Why the 2D reduction is the right next step	44
34.2	2D governing equation used as the Module 4 target	44
34.3	Source terms in the radiation-effects setting	44
34.4	How ballistic terms should be interpreted in this project	45
34.5	Coupling to Module 3 defect evolution	45
34.6	Coupling to Modules 5 and 6	45
34.7	Verification logic for the revised Module 4	45
34.8	Summary of the revised Module 4 role	46
35	References	46

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
BTE	Boltzmann transport equation	Phase-space transport equation for heat carriers such as phonons, electrons, or molecules.
BD	Ballistic-diffusive	Reduced model in which boundary-originating carriers are treated ballistically and internally scattered carriers are treated diffusively.
FEM	Finite element method	Numerical method based on a weak form, basis functions, and element-wise integration.
PDE	Partial differential equation	Equation involving partial derivatives with respect to space and/or time.
RTA	Relaxation-time approximation	Collision model in which the distribution relaxes toward local equilibrium over a characteristic time.
P1 approximation	First-order spherical harmonics approximation	Angular-moment closure that keeps the isotropic part and first directional anisotropy of a distribution.
t	Time	Independent time coordinate.
$\mathbf{r}$	Position vector	Spatial coordinate in the material domain.
x	One-dimensional coordinate	Spatial coordinate across a thin film.
L	Characteristic length	Film thickness or representative device length scale.
s	Ray coordinate	Distance measured along a carrier propagation direction.
s_0	Boundary ray coordinate	Ray coordinate at the boundary point from which a carrier originates.
$\hat{\Omega}$	Propagation direction	Unit vector in the direction of carrier motion.
$\mathbf{v}$	Carrier group velocity	Velocity vector of a carrier wave packet.
v	Carrier speed	Magnitude of the carrier group velocity, $v = \mathbf{v} $.
μ	Direction cosine	In one-dimensional transport, $\mu = \cos \theta$, where θ is the angle relative to the positive x -axis.
ω	Angular frequency	Carrier angular frequency.

Symbol / acronym	Name	Meaning in this document
$\hbar$	Reduced Planck constant	Quantum of action divided by 2π .
$D(\omega)$	Density of states	Number of available carrier modes per unit volume per angular-frequency interval.
f	Distribution function	Carrier occupation in phase space.
f_0	Local-equilibrium distribution	Bose-Einstein or Fermi-Dirac distribution determined by local thermodynamic variables.
f_b	Ballistic distribution	Boundary-originating portion of f .
f_m	Medium/scattered distribution	Portion of f associated with carriers scattered or emitted inside the medium.
I_ω	Spectral intensity	Directional energy-flow representation of the distribution function.
$I_{0\omega}$	Equilibrium spectral intensity	Intensity corresponding to local equilibrium.
$I_{b\omega}$	Ballistic intensity	Boundary-originating part of I_ω .
$I_{m\omega}$	Medium/scattered intensity	Internally scattered or emitted part of I_ω .
$I_{w\omega}$	Wall/boundary intensity	Intensity emitted or reflected from a boundary into the domain.
$J_{0\omega}$	Isotropic intensity coefficient	Zeroth angular harmonic of $I_{m\omega}$.
$\mathbf{J}_{1\omega}$	First angular coefficient	First angular harmonic vector of $I_{m\omega}$.
τ_ω	Mode-dependent relaxation time	Average time between scattering events for carriers of frequency ω .
τ	Averaged relaxation time	Spectrally averaged relaxation time used in the reduced model.
Λ_ω	Mode-dependent mean free path	$\Lambda_\omega = v\tau_\omega$.
Λ	Mean free path	Average distance traveled between scattering events.
Kn	Knudsen number	$\text{Kn} = \Lambda/L$, ratio of mean free path to device length.
u	Internal energy density	Carrier energy per unit volume.
u_b	Ballistic internal energy density	Internal energy density associated with $I_{b\omega}$.
u_m	Medium/scattered internal energy density	Internal energy density associated with $I_{m\omega}$.

Symbol / acronym	Name	Meaning in this document
u_{b0}	Equilibrium ballistic energy density	Ballistic energy contribution in a reference equilibrium state.
u_{m0}	Equilibrium medium energy density	Medium energy contribution in a reference equilibrium state.
T	Equivalent temperature	Temperature inferred from internal energy; a true thermodynamic temperature only when local equilibrium is valid.
T_m	Medium-equivalent temperature	$T_m = u_m/C$ when constant volumetric heat capacity is assumed.
T_b	Ballistic-equivalent temperature	$T_b = u_b/C$ when constant volumetric heat capacity is assumed.
T_0	Initial/reference temperature	Uniform initial temperature of a validation film.
T_1	Emitted boundary temperature	Temperature characterizing carriers emitted from the hot boundary.
ΔT	Temperature scale	$\Delta T = T_1 - T_0$.
C	Volumetric heat capacity	$C = \partial u / \partial T$, energy per unit volume per kelvin.
C_ω	Spectral heat capacity	Heat capacity contribution from carriers near angular frequency ω .
k	Thermal conductivity	Diffusive thermal conductivity.
α	Thermal diffusivity	$\alpha = k/C$.
$\mathbf{q}$	Total heat flux	Energy flow per unit area per unit time.
$\mathbf{q}_b$	Ballistic heat flux	Heat flux carried by boundary-originating carriers.
$\mathbf{q}_m$	Medium/scattered heat flux	Heat flux carried by internally scattered carriers.
$\dot{q}_e$	Volumetric heat generation	External heat source per unit volume.
$\mathbf{n}$	Unit normal	Outward unit normal vector on a boundary.
Ω	Spatial domain	Physical region occupied by the simulated material.
$\partial\Omega$	Boundary of the domain	Surface enclosing Ω .
w	Test function	Weighting function used to form the FEM weak statement.
N_i	FEM basis function	Shape function associated with node i .
M	Mass matrix	FEM matrix from $\int_\Omega N_i N_j dV$.

Symbol / acronym	Name	Meaning in this document
K	Stiffness matrix	FEM matrix from $\int_{\Omega} \nabla N_i \cdot \alpha \nabla N_j dV$.
M_C	Heat-capacity mass matrix	FEM matrix $\int_{\Omega} C N_i N_j dA$ used in the temperature form of Module 4.
K_T	Thermal conductivity matrix	FEM matrix $\int_{\Omega} k \nabla N_i \cdot \nabla N_j dA$.
$\mathbf{T}$	Nodal temperature vector	Vector of nodal values used to represent the FEM temperature field.
$\mathbf{V}$	Nodal temperature-rate vector	Vector of nodal values approximating $\partial T / \partial t$.
Δt	Time step	Time increment used in the backward-Euler FEM update.
Ω_e	Element domain	Area occupied by one triangular finite element.
A_e	Element area	Physical area of one triangular element.
η	Nondimensional coordinate	$\eta = x/L$.
t^*	Nondimensional time	$t^* = t/\tau$.
θ_m	Nondimensional medium energy	Normalized version of $u_m - u_{m0}$.
θ_b	Nondimensional ballistic energy	Normalized version of $u_b - u_{b0}$.
θ	Nondimensional total energy temperature	Normalized total energy perturbation, $\theta = \theta_m + \theta_b$.
q_{b0}	Equilibrium ballistic heat flux	Reference ballistic flux subtracted in the normalized validation problem.
q_b^*	Nondimensional ballistic flux	Normalized version of $q_b - q_{b0}$.
$E_n(z)$	Exponential integral	Integral $E_n(z) = \int_1^{\infty} e^{-zt} t^{-n} dt$, useful for equilibrium ballistic terms.

1 What problem this module is solving

A superconducting qubit chip contains structures whose relevant dimensions can range from millimeters, to micrometers, to nanometers. A local energy deposition event, a photon absorption event, or a transient heat pulse does not always relax according to ordinary macroscopic Fourier conduction. At small length scales, heat carriers can travel a significant fraction of the device size

before scattering. In that case, the heat flux at a point is partly determined by distant boundaries and by the history of carrier flight, not only by the local temperature gradient.

The purpose of this module is to construct a thermal transport model that is more physical than Fourier conduction but much cheaper than a full Boltzmann transport solve. The model should be capable of representing the transition between these limits:

$$\text{diffusive limit: } \Lambda \ll L, \quad (1.1)$$

$$\text{quasi-ballistic limit: } \Lambda \sim L, \quad (1.2)$$

$$\text{ballistic limit: } \Lambda \gtrsim L, \quad (1.3)$$

where Λ is the carrier mean free path and L is the characteristic device length.

Core intuition. Fourier's law is a local law: the heat flux at $\mathbf{r}$ is determined by $\nabla T(\mathbf{r})$. Ballistic transport is nonlocal: a carrier at $\mathbf{r}$ may have originated at a boundary point many mean-free-path-scale distances away and may remember the boundary state at an earlier time.

2 Heat carriers and the meaning of temperature

2.1 Heat is carried by microscopic excitations

In a solid, thermal energy is transported by microscopic excitations. In an insulating crystal, phonons are usually the dominant heat carriers. In a metal, both electrons and phonons can contribute. In a superconducting qubit material stack, the situation becomes more complicated at cryogenic temperature because phonons, quasiparticles, Cooper pairs, normal-metal traps, and interfaces can all matter. Module 4 intentionally starts with the room-temperature silicon heat-transport problem because it provides the transport language needed later in the radiation-effects framework.

A carrier is described by its position $\mathbf{r}$, propagation direction $\hat{\Omega}$, frequency ω , and time t . A distribution function tells us how many carriers occupy a small region of this phase space.

2.2 Temperature is simple only near local equilibrium

Macroscopic heat conduction often begins by assuming that every small volume element is in local thermal equilibrium. Then one can assign a temperature $T(\mathbf{r}, t)$, and the local internal energy density is a function of temperature:

$$u(\mathbf{r}, t) = u[T(\mathbf{r}, t)]. \quad (2.1)$$

For small temperature changes around a reference temperature, this becomes

$$du = C dT, \quad (2.2)$$

where C is volumetric heat capacity. This relation is useful but should not be mistaken for a definition that is always valid. If the carrier distribution is strongly nonequilibrium, a single thermodynamic temperature may not exist.

In this module we use *equivalent temperature* in a precise sense: it is the temperature that an equilibrium distribution would need to have in order to contain the same local internal energy. Therefore, in ballistic transport, temperature is best interpreted as an energy label, not always as a true local-equilibrium state variable.

3 Macroscopic energy conservation

Before discussing any constitutive law, we start with energy conservation. Consider an arbitrary fixed control volume V with boundary ∂V and outward normal $\mathbf{n}$. Let u be internal energy density and $\mathbf{q}$ be heat flux. Conservation of energy says

$$\frac{d}{dt} \int_V u dV = - \int_{\partial V} \mathbf{q} \cdot \mathbf{n} dA + \int_V \dot{q}_e dV, \quad (3.1)$$

where $\dot{q}_e$ is a volumetric heat source. The first term on the right is energy leaving the volume by heat flow. Using the divergence theorem,

$$\int_{\partial V} \mathbf{q} \cdot \mathbf{n} dA = \int_V \nabla \cdot \mathbf{q} dV. \quad (3.2)$$

Because the control volume is arbitrary,

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{q} = \dot{q}_e.} \quad (3.3)$$

This equation is exact at the continuum energy-balance level. The modeling question is how to compute $\mathbf{q}$.

4 Fourier conduction as a local constitutive model

Fourier conduction assumes

$$\boxed{\mathbf{q} = -k \nabla T.} \quad (4.1)$$

The minus sign means heat flows from high temperature to low temperature. If $u = CT$ with constant C , substitution into Eq. (3.3) gives

$$C \frac{\partial T}{\partial t} + \nabla \cdot (-k \nabla T) = \dot{q}_e, \quad (4.2)$$

so

$$\boxed{C \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{q}_e.} \quad (4.3)$$

If k is constant and there is no source,

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T, \quad \alpha = \frac{k}{C}. \quad (4.4)$$

4.1 Why the Fourier heat equation has infinite propagation speed

For an infinite one-dimensional medium, the Green's function of

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad (4.5)$$

from a point impulse at $t = 0$ is

$$G(x, t) = \frac{1}{\sqrt{4\pi\alpha t}} \exp\left(-\frac{x^2}{4\alpha t}\right), \quad t > 0. \quad (4.6)$$

For every $t > 0$, the exponential is nonzero for every finite x . The disturbance immediately has nonzero amplitude arbitrarily far away. Mathematically, this is because the heat equation is parabolic. Physically, this is incompatible with heat being carried by microscopic carriers moving at finite group velocity.

At macroscale, this infinite-speed artifact is usually harmless. At nanoscale, microscale, or ultrafast timescales, it can dominate early-time predictions such as surface heat flux.

5 The Knudsen number and transport regimes

The mean free path Λ is the average distance a carrier travels before scattering. The relaxation time τ is the average time between scattering events. If the carrier speed is v , then

$$\boxed{\Lambda = v\tau.} \quad (5.1)$$

The Knudsen number is

$$\boxed{\text{Kn} = \frac{\Lambda}{L}.} \quad (5.2)$$

It compares the microscopic distance between scattering events with the macroscopic device length.

Regime	Scaling	Physical interpretation
Diffusive	$\text{Kn} \ll 1$	Carriers scatter many times before crossing the system. Local equilibrium is a good approximation. Fourier conduction is usually valid.
Quasi-ballistic	$\text{Kn} \sim 0.1 - 1$	A significant fraction of carriers travel across device-scale distances before scattering. Boundary memory and finite flight time matter.
Ballistic	$\text{Kn} \gtrsim 1$	Carriers can cross the domain with few or no scattering events. Pure diffusion is a poor approximation.

6 Microscopic starting point: distribution function

Let $f(t, \mathbf{r}, \mathbf{k})$ be the carrier distribution function, where $\mathbf{k}$ is wave vector. For a phonon branch, the angular frequency is $\omega = \omega(\mathbf{k})$, and the group velocity is

$$\mathbf{v} = \nabla_{\mathbf{k}} \omega(\mathbf{k}). \quad (6.1)$$

The quantity f is an occupation number: it tells us how many carriers occupy a phase-space state. For phonons at local equilibrium,

$$f_0(\omega, T) = \frac{1}{\exp(\hbar\omega/k_{\text{B}}T) - 1}, \quad (6.2)$$

which is the Bose-Einstein distribution. For electrons, the equilibrium distribution would be Fermi-Dirac. This derivation focuses on the phonon-like heat-carrier picture, but the transport structure is more general.

7 Deriving the Boltzmann transport equation

7.1 Streaming without scattering

First ignore collisions. If carriers move with velocity $\mathbf{v}$, then the distribution is conserved along a trajectory in phase space. In the absence of external force, the wave vector is constant and

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{d\mathbf{r}}{dt} \cdot \nabla_{\mathbf{r}} f = \frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f. \quad (7.1)$$

Conservation along trajectories gives

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f = 0. \quad (7.2)$$

This is free streaming.

7.2 Adding scattering

Scattering changes the occupation of a state. The general Boltzmann equation is

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}}, \quad (7.3)$$

where the right side is the collision operator. The exact collision operator can be complicated because it can include phonon-phonon scattering, phonon-boundary scattering, impurity scattering, electron-phonon scattering, and other mechanisms.

The relaxation-time approximation replaces the detailed collision operator by a tendency to relax toward local equilibrium:

$$\left(\frac{\partial f}{\partial t} \right)_{\text{coll}} = -\frac{f - f_0}{\tau_{\omega}}. \quad (7.4)$$

Therefore,

$$\boxed{\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f = -\frac{f - f_0}{\tau_{\omega}}}. \quad (7.5)$$

This is the Boltzmann transport equation under the relaxation-time approximation.

7.3 Term-by-term meaning

Each term in Eq. (7.5) has a direct physical meaning:

$$\frac{\partial f}{\partial t} : \text{local time change of carrier occupation}, \quad (7.6)$$

$$\mathbf{v} \cdot \nabla_{\mathbf{r}} f : \text{change due to carriers streaming through space}, \quad (7.7)$$

$$-\frac{f - f_0}{\tau_{\omega}} : \text{relaxation toward local equilibrium by scattering}. \quad (7.8)$$

If $f > f_0$, scattering reduces f . If $f < f_0$, scattering increases f . The relaxation time τ_ω can depend on frequency because different phonon modes scatter differently.

Assumption in this module. We neglect explicit force terms. For electron transport in an electric field, the BTE would include a force term in wave-vector space. For phonon heat transport without external body force, the streaming-plus-collision form is the relevant starting point.

8 Why a direct Boltzmann solve is expensive

In three-dimensional space, f depends on

$$(x, y, z), \quad \text{two direction angles}, \quad \omega, \quad t. \quad (8.1)$$

Equivalently, the BTE lives in real space, momentum space, and time. Solving it directly is much more expensive than solving a heat equation for $T(x, y, z, t)$. The purpose of the ballistic-diffusive approximation is to keep the finite-speed and nonlocal physics that matter in small structures while reducing the problem to a PDE in ordinary space and time plus an explicitly computed ballistic flux term.

9 Intensity notation

9.1 Definition of spectral intensity

It is useful to rewrite the BTE in terms of spectral intensity. For an isotropic medium, define

$$I_\omega(t, \mathbf{r}, \hat{\Omega}) = \frac{1}{4\pi} v \hbar \omega D(\omega) f(t, \mathbf{r}, \hat{\Omega}, \omega), \quad (9.1)$$

where $D(\omega)$ is the density of states per unit volume. The factor $1/(4\pi)$ distributes the isotropic density over solid angle. The exact convention can vary, but this convention makes the angular moments transparent.

The intensity form of the BTE is

$$\boxed{\frac{\partial I_\omega}{\partial t} + \mathbf{v} \cdot \nabla I_\omega = -\frac{I_\omega - I_{0\omega}}{\tau_\omega}.} \quad (9.2)$$

Here $I_{0\omega}$ is the local-equilibrium intensity.

9.2 Moments of intensity

The internal energy density is obtained by integrating energy per unit volume over frequency and direction. Because intensity is energy flow per area per time per solid angle per frequency, division by speed converts flux-like intensity into energy density:

$$u(t, \mathbf{r}) = \int_0^\infty \int_{4\pi} \frac{I_\omega(t, \mathbf{r}, \hat{\Omega})}{v} d\Omega d\omega. \quad (9.3)$$

The heat flux is the directional projection of energy flow:

$$\mathbf{q}(t, \mathbf{r}) = \int_0^\infty \int_{4\pi} I_\omega(t, \mathbf{r}, \hat{\Omega}) \hat{\Omega} d\Omega d\omega. \quad (9.4)$$

These two moment definitions are the bridge between microscopic carrier transport and continuum thermal variables.

10 Recovering Fourier conduction from the BTE

This section is important because it shows what Fourier conduction throws away. Suppose the system is close to local equilibrium, so

$$f = f_0 + f_1, \quad |f_1| \ll |f_0|. \quad (10.1)$$

Assume steady or slowly varying conditions so the leading nonequilibrium correction comes from spatial gradients. The BTE gives approximately

$$\mathbf{v} \cdot \nabla f_0 \approx -\frac{f_1}{\tau_\omega}, \quad (10.2)$$

so

$$f_1 \approx -\tau_\omega \mathbf{v} \cdot \nabla f_0. \quad (10.3)$$

Because $f_0 = f_0[T(\mathbf{r})]$,

$$\nabla f_0 = \frac{\partial f_0}{\partial T} \nabla T. \quad (10.4)$$

Thus

$$f_1 \approx -\tau_\omega (\mathbf{v} \cdot \nabla T) \frac{\partial f_0}{\partial T}. \quad (10.5)$$

The heat flux is the first velocity moment of the nonequilibrium correction. In an isotropic medium,

$$\int v_i v_j d\Omega = \frac{4\pi}{3} v^2 \delta_{ij}. \quad (10.6)$$

Therefore the flux becomes

$$\mathbf{q} = -\frac{1}{3} \int_0^\infty C_\omega v^2 \tau_\omega d\omega \nabla T. \quad (10.7)$$

Using $\Lambda_\omega = v\tau_\omega$, this is

$$\mathbf{q} = -k \nabla T, \quad k = \frac{1}{3} \int_0^\infty C_\omega v \Lambda_\omega d\omega. \quad (10.8)$$

This is Fourier's law from kinetic theory.

What had to be assumed to get Fourier's law? We assumed near local equilibrium, small anisotropy, many scattering events over the temperature-gradient length scale, and no strong boundary memory. Those assumptions are precisely what become questionable when Kn is not small.

11 Cattaneo's correction and its limitation

11.1 Flux relaxation

The Cattaneo model modifies Fourier's law by giving heat flux a relaxation time:

$$\tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -k \nabla T. \quad (11.1)$$

If $\tau \rightarrow 0$, Fourier's law is recovered. Combining Eq. (11.1) with energy conservation gives a hyperbolic heat equation. For constant C , constant k , and no source,

$$C \frac{\partial T}{\partial t} + \nabla \cdot \mathbf{q} = 0. \quad (11.2)$$

Take a time derivative:

$$C \frac{\partial^2 T}{\partial t^2} + \nabla \cdot \frac{\partial \mathbf{q}}{\partial t} = 0. \quad (11.3)$$

Take the divergence of Eq. (11.1):

$$\tau \nabla \cdot \frac{\partial \mathbf{q}}{\partial t} + \nabla \cdot \mathbf{q} = -k \nabla^2 T. \quad (11.4)$$

Using $\nabla \cdot \mathbf{q} = -C \partial T / \partial t$, we get

$$\tau \frac{\partial^2 T}{\partial t^2} + \frac{\partial T}{\partial t} = \alpha \nabla^2 T. \quad (11.5)$$

This is a telegraph-type equation. It has finite signal speed

$$c_{\text{th}} = \sqrt{\frac{\alpha}{\tau}}. \quad (11.6)$$

Using $\alpha = k/C$ and the kinetic estimate $k \approx Cv\Lambda/3 = Cv^2\tau/3$,

$$c_{\text{th}} \approx \frac{v}{\sqrt{3}}. \quad (11.7)$$

So Cattaneo fixes the most obvious mathematical pathology of Fourier conduction: infinite propagation speed.

11.2 Why Cattaneo is still not enough

Cattaneo still treats the entire heat flux as a local constitutive variable that relaxes toward $-k\nabla T$. It does not explicitly represent carriers that left a boundary, crossed the domain, and arrived at an interior point without scattering. Therefore, it does not contain the correct nonlocal boundary memory.

This missing ingredient matters in thin films and nanoscale structures. In a heating experiment where one boundary suddenly emits carriers, some carriers travel ballistically into the film before enough scattering has occurred to establish a local gradient-driven flux. Cattaneo can produce wave-like behavior, but that wave is not the same as a distribution of particles arriving from the boundary at retarded times. In Chen's comparison calculations, Cattaneo gives artificial surface heat-flux oscillations and poor agreement with the Boltzmann equation in the small-scale transient regime.

12 Ballistic-diffusive idea

The ballistic-diffusive model starts from a simple but powerful split:

$$I_\omega = I_{b\omega} + I_{m\omega}. \quad (12.1)$$

The two parts have different physical meanings:

$$I_{b\omega} : \text{carriers that originated at a boundary and have not yet scattered,} \quad (12.2)$$

$$I_{m\omega} : \text{carriers produced by scattering or emission inside the medium.} \quad (12.3)$$

Why the split is useful. The ballistic part is highly directional and nonlocal, so we should not approximate it by diffusion. The medium part has already undergone scattering and is more nearly isotropic, so a diffusion-like angular closure is more defensible.

This is not merely a mathematical trick. It corresponds to two populations of heat carriers with different histories:

1. A boundary-originating population that remembers the boundary condition and flight time.
2. An internally scattered population that has partially lost directional memory.

13 Equation for the ballistic component

The ballistic component experiences streaming and out-scattering, but it has no in-scattering source because newly scattered carriers are assigned to $I_{m\omega}$. Therefore,

$$\boxed{\frac{1}{v} \frac{\partial I_{b\omega}}{\partial t} + \hat{\Omega} \cdot \nabla I_{b\omega} = -\frac{I_{b\omega}}{v\tau_{\omega}} = -\frac{I_{b\omega}}{\Lambda_{\omega}}.} \quad (13.1)$$

13.1 Derivation along a characteristic

Let s be distance along a ray in direction $\hat{\Omega}$. Along the ray,

$$\mathbf{r}(s) = \mathbf{r}_0 + s\hat{\Omega}. \quad (13.2)$$

For a time-dependent problem, a carrier travels a distance ds in time $dt = ds/v$. Along the characteristic,

$$\frac{dI_{b\omega}}{ds} = \frac{1}{v} \frac{\partial I_{b\omega}}{\partial t} + \hat{\Omega} \cdot \nabla I_{b\omega}. \quad (13.3)$$

Therefore Eq. (13.1) becomes

$$\frac{dI_{b\omega}}{ds} = -\frac{I_{b\omega}}{\Lambda_{\omega}}. \quad (13.4)$$

The solution is

$$I_{b\omega}(s) = I_{b\omega}(s_0) \exp \left[-\int_{s_0}^s \frac{ds'}{\Lambda_{\omega}(s')} \right]. \quad (13.5)$$

For a homogeneous medium, this reduces to

$$I_{b\omega}(s) = I_{b\omega}(s_0) \exp \left[-\frac{s - s_0}{\Lambda_{\omega}} \right]. \quad (13.6)$$

The exponential is a survival probability: it is the probability that a carrier travels from s_0 to s without scattering.

13.2 Retarded boundary time

If the boundary intensity changes with time, the carrier arriving at $(t, \mathbf{r})$ did not leave the boundary at time t . It left earlier by the flight time

$$\Delta t_{\text{flight}} = \frac{s - s_0}{v}. \quad (13.7)$$

Thus the ballistic solution is

$$I_{b\omega}(t, \mathbf{r}, \hat{\Omega}) = I_{w\omega} \left(t - \frac{s - s_0}{v}, \mathbf{r} - (s - s_0)\hat{\Omega} \right) \exp \left[- \int_{s_0}^s \frac{ds'}{\Lambda_{\omega}(s')} \right]. \quad (13.8)$$

This is one of the most important equations in the module. It contains both finite propagation speed and attenuation by scattering.

14 Equation for the medium/scattered component

The remaining carriers are assigned to the medium component. They obey

$$\frac{\partial I_{m\omega}}{\partial t} + \mathbf{v} \cdot \nabla I_{m\omega} = - \frac{I_{m\omega} - I_{0\omega}}{\tau_{\omega}}. \quad (14.1)$$

This equation still resembles the BTE, but now the distribution is more isotropic than the full distribution because the strongly directional unscattered boundary population has been separated out.

15 P1 angular expansion for the medium component

The diffusion approximation is an angular-moment closure. We expand $I_{m\omega}$ in angular harmonics and keep only the isotropic term and the first directional term:

$$I_{m\omega}(t, \mathbf{r}, \hat{\Omega}) \approx J_{0\omega}(t, \mathbf{r}) + \mathbf{J}_{1\omega}(t, \mathbf{r}) \cdot \hat{\Omega}. \quad (15.1)$$

The scalar $J_{0\omega}$ represents the angular average. The vector $\mathbf{J}_{1\omega}$ represents the leading directional bias.

The angular identities needed are

$$\int_{4\pi} d\Omega = 4\pi, \quad (15.2)$$

$$\int_{4\pi} \hat{\Omega} d\Omega = \mathbf{0}, \quad (15.3)$$

$$\int_{4\pi} \hat{\Omega}_i \hat{\Omega}_j d\Omega = \frac{4\pi}{3} \delta_{ij}. \quad (15.4)$$

These identities encode isotropy of the sphere. The factor $1/3$ is the same factor that appears in kinetic theory expressions such as $k = (1/3)Cv\Lambda$.

16 Moment equations for the medium component

Substitute Eq. (15.1) into Eq. (14.1). Since $\mathbf{v} = v\hat{\Omega}$,

$$\frac{\partial J_{0\omega}}{\partial t} + \frac{\partial \mathbf{J}_{1\omega}}{\partial t} \cdot \hat{\Omega} + v\hat{\Omega} \cdot \nabla J_{0\omega} + v\hat{\Omega} \cdot \nabla (\mathbf{J}_{1\omega} \cdot \hat{\Omega}) = -\frac{J_{0\omega} - I_{0\omega}}{\tau_\omega} - \frac{\mathbf{J}_{1\omega} \cdot \hat{\Omega}}{\tau_\omega}. \quad (16.1)$$

16.1 Zeroth angular moment

Integrate Eq. (16.1) over all directions. Terms odd in $\hat{\Omega}$ vanish. Using Eq. (15.4),

$$4\pi \frac{\partial J_{0\omega}}{\partial t} + \frac{4\pi}{3} v \nabla \cdot \mathbf{J}_{1\omega} = -\frac{4\pi}{\tau_\omega} (J_{0\omega} - I_{0\omega}). \quad (16.2)$$

Equivalently,

$$\frac{1}{v} \frac{\partial J_{0\omega}}{\partial t} + \frac{1}{3} \nabla \cdot \mathbf{J}_{1\omega} = -\frac{J_{0\omega} - I_{0\omega}}{v\tau_\omega} = -\frac{J_{0\omega} - I_{0\omega}}{\Lambda_\omega}. \quad (16.3)$$

16.2 First angular moment

Multiply Eq. (16.1) by $\hat{\Omega}$ and integrate over all directions. The result is

$$\frac{1}{v} \frac{\partial \mathbf{J}_{1\omega}}{\partial t} + \nabla J_{0\omega} = -\frac{\mathbf{J}_{1\omega}}{\Lambda_\omega}. \quad (16.4)$$

This equation says that the directional bias $\mathbf{J}_{1\omega}$ relaxes over a mean free path and is driven by gradients of the angular average $J_{0\omega}$.

17 Heat flux and internal energy split

Using the intensity moments, the total internal energy density is

$$u = u_b + u_m, \quad (17.1)$$

where

$$u_b = \int_0^\infty \int_{4\pi} \frac{I_{b\omega}}{v} d\Omega d\omega, \quad (17.2)$$

$$u_m = \int_0^\infty \int_{4\pi} \frac{I_{m\omega}}{v} d\Omega d\omega. \quad (17.3)$$

With the P1 expansion,

$$u_m = \int_0^\infty \frac{4\pi J_{0\omega}}{v} d\omega. \quad (17.4)$$

Similarly, the total heat flux is

$$\mathbf{q} = \mathbf{q}_b + \mathbf{q}_m, \quad (17.5)$$

where

$$\mathbf{q}_b = \int_0^\infty \int_{4\pi} I_{b\omega} \hat{\Omega} \, d\Omega \, d\omega, \quad (17.6)$$

$$\mathbf{q}_m = \int_0^\infty \int_{4\pi} I_{m\omega} \hat{\Omega} \, d\Omega \, d\omega. \quad (17.7)$$

Using the P1 expansion and angular identities,

$$\mathbf{q}_m = \frac{4\pi}{3} \int_0^\infty \mathbf{J}_{1\omega} \, d\omega. \quad (17.8)$$

18 Constitutive relation for the medium component

Start from the first moment equation,

$$\frac{1}{v} \frac{\partial \mathbf{J}_{1\omega}}{\partial t} + \nabla J_{0\omega} = -\frac{\mathbf{J}_{1\omega}}{\Lambda_\omega}. \quad (18.1)$$

Multiply by the spectral weight that converts $\mathbf{J}_{1\omega}$ into heat flux and integrate over frequency. Under the standard averaged-parameter approximation, the result is

$$\tau \frac{\partial \mathbf{q}_m}{\partial t} + \mathbf{q}_m = -\frac{k}{C} \nabla u_m. \quad (18.2)$$

If we define $T_m = u_m/C$, then

$$\tau \frac{\partial \mathbf{q}_m}{\partial t} + \mathbf{q}_m = -k \nabla T_m. \quad (18.3)$$

This looks like Cattaneo's law, but the interpretation is different. In Cattaneo theory, $\mathbf{q}$ is the total heat flux. Here, $\mathbf{q}_m$ is only the scattered medium flux. The ballistic flux $\mathbf{q}_b$ is not represented by $-k \nabla T_m$; it is computed separately from the boundary-originating carrier population.

19 Energy equation for the medium component

Use total energy conservation,

$$\frac{\partial(u_m + u_b)}{\partial t} + \nabla \cdot (\mathbf{q}_m + \mathbf{q}_b) = \dot{q}_e. \quad (19.1)$$

Rearrange to isolate the medium component:

$$\frac{\partial u_m}{\partial t} + \nabla \cdot \mathbf{q}_m = \dot{q}_e - \frac{\partial u_b}{\partial t} - \nabla \cdot \mathbf{q}_b. \quad (19.2)$$

The right side says that the medium energy changes not only due to the external source, but also due to energy removed from or added by the ballistic component.

19.1 Ballistic moment relation

The ballistic intensity equation implies a moment equation. Integrating Eq. (13.1) over direction and frequency gives

$$\frac{\partial u_b}{\partial t} + \nabla \cdot \mathbf{q}_b = -\frac{u_b}{\tau} \quad (19.3)$$

for an averaged relaxation-time model. Equivalently,

$$\tau \frac{\partial u_b}{\partial t} + \tau \nabla \cdot \mathbf{q}_b = -u_b. \quad (19.4)$$

This states that the ballistic energy population is depleted by scattering over time τ . Once a ballistic carrier scatters, it leaves the ballistic population and becomes part of the medium population.

19.2 Eliminating the medium heat flux

Take the divergence of Eq. (18.2):

$$\tau \frac{\partial}{\partial t} (\nabla \cdot \mathbf{q}_m) + \nabla \cdot \mathbf{q}_m = -\nabla \cdot \left(\frac{k}{C} \nabla u_m \right). \quad (19.5)$$

From Eq. (19.2),

$$\nabla \cdot \mathbf{q}_m = \dot{q}_e - \frac{\partial u_m}{\partial t} - \frac{\partial u_b}{\partial t} - \nabla \cdot \mathbf{q}_b. \quad (19.6)$$

Substitute Eq. (19.6) into Eq. (19.5). After collecting terms,

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot \left(\frac{k}{C} \nabla u_m \right) + \dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} - \nabla \cdot \mathbf{q}_b \quad (19.7)$$

$$- \left[\tau \frac{\partial^2 u_b}{\partial t^2} + \frac{\partial u_b}{\partial t} + \tau \frac{\partial}{\partial t} (\nabla \cdot \mathbf{q}_b) \right]. \quad (19.8)$$

The bracketed term vanishes by differentiating Eq. (19.4). Therefore,

$$\boxed{\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot \left(\frac{k}{C} \nabla u_m \right) - \nabla \cdot \mathbf{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right)}. \quad (19.9)$$

This is the ballistic-diffusive heat-conduction equation in energy-density form.

Interpretation of the final equation. The left side is a Cattaneo-like inertial thermal response of the scattered medium energy. The first term on the right is diffusion of the scattered medium energy. The second term, $-\nabla \cdot \mathbf{q}_b$, is the explicit source/sink caused by divergence of the boundary-originating ballistic heat flux. The final terms represent external volumetric heating and its relaxation-time correction.

20 Explicit ballistic heat flux

The ballistic flux is obtained by inserting the characteristic solution Eq. (13.8) into the flux moment:

$$\mathbf{q}_b(t, \mathbf{r}) = \int_0^\infty \int_{4\pi} I_{w\omega} \left(t - \frac{s - s_0}{v}, \mathbf{r} - (s - s_0)\hat{\Omega} \right) \exp \left[- \int_{s_0}^s \frac{ds'}{\Lambda_\omega(s')} \right] \hat{\Omega} d\Omega d\omega. \quad (20.1)$$

This term depends on boundary emission/reflection, flight time, geometry, and attenuation. In simple geometries, it can be evaluated analytically or semi-analytically. In complex 3D geometries, it may be approximated by ray tracing, angular quadrature, or reduced boundary-source models.

21 Boundary condition for the medium component

Because boundary-originating carriers are assigned to the ballistic part, the medium component at a boundary consists only of carriers incident on the boundary from inside the material. This leads to a Marshak-type boundary condition.

At a boundary with outward normal $\mathbf{n}$, the incident medium flux through the boundary is determined by directions satisfying $\hat{\Omega} \cdot \mathbf{n} < 0$. The P1 approximation gives

$$\mathbf{q}_m \cdot \mathbf{n} = -\frac{v}{2} u_m \quad (21.1)$$

for a black-wall limiting form. Combining this with the medium constitutive relation gives

$$\tau \frac{\partial u_m}{\partial t} + u_m = -\frac{\Lambda}{3} \nabla u_m \cdot \mathbf{n}. \quad (21.2)$$

This is not the same as prescribing the boundary temperature in Fourier conduction. It is a boundary condition for the scattered medium energy after boundary-originating carriers have been assigned to the ballistic component.

22 Temperature consistency at boundaries

The boundary temperature that characterizes emitted carriers is not always identical to the equivalent equilibrium temperature just inside the medium. This distinction is central in nanoscale transport. Suppose a boundary emits phonons as if it were at temperature T_1 . The incoming carrier population is then specified by an emitted distribution. Inside the film, however, the total local energy is

$$u(x, t) = u_m(x, t) + u_b(x, t). \quad (22.1)$$

An equivalent internal-energy temperature would be

$$T_{\text{eq}}(x, t) = T_0 + \frac{u(x, t) - u_0}{C} \quad (22.2)$$

for a constant heat capacity model. In a highly nonequilibrium boundary layer, T_{eq} need not equal the emitted boundary temperature. Therefore, a naively imposed Dirichlet temperature boundary condition can be physically inconsistent.

This is why the ballistic-diffusive formulation keeps the boundary-emitted ballistic population separate from the internal medium population. It avoids forcing all boundary physics into a local equilibrium temperature.

23 One-dimensional thin-film validation problem

23.1 Why this validation problem is the right first test

The canonical validation problem is transient heat conduction across a film of thickness L . The coordinate is $x \in [0, L]$. The film starts at uniform reference temperature T_0 . At $t = 0$, the left boundary begins emitting carriers characterized by a higher emitted temperature T_1 , while the right boundary remains characterized by T_0 . There is no volumetric heating:

$$\dot{q}_e = 0. \quad (23.1)$$

This problem is simple geometrically but difficult physically. It contains all of the effects that distinguish Fourier, Cattaneo, Boltzmann, and ballistic-diffusive descriptions:

1. a finite device length L ,
2. a finite carrier mean free path Λ ,
3. a finite carrier flight time across the film,
4. a boundary-emitted carrier population,

5. a scattered carrier population inside the film,
6. a temperature or energy jump issue at the boundary.

Slide takeaway. The thin-film problem is the cleanest benchmark because it is one-dimensional in space but still contains the real nanoscale physics: finite mean free path, finite carrier speed, retarded boundary emission, and boundary temperature subtlety.

23.2 Starting from the full ballistic-diffusive energy equation

The general ballistic-diffusive medium-energy equation derived above is

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot \left(\frac{k}{C} \nabla u_m \right) - \nabla \cdot \mathbf{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right). \quad (23.2)$$

The unknown in this equation is not the total internal energy u , but the medium or scattered-carrier energy u_m . The ballistic energy u_b and ballistic flux $\mathbf{q}_b$ are computed from boundary-originating carriers. The total energy is reconstructed afterward:

$$u(x, t) = u_m(x, t) + u_b(x, t). \quad (23.3)$$

For the one-dimensional thin film, all spatial variation is along x , and the ballistic flux has only an x -component:

$$\mathbf{q}_b(x, t) = q_b(x, t) \hat{x}. \quad (23.4)$$

Therefore,

$$\nabla \cdot \mathbf{q}_b = \frac{\partial q_b}{\partial x}, \quad \nabla \cdot \left(\frac{k}{C} \nabla u_m \right) = \frac{\partial}{\partial x} \left(\frac{k}{C} \frac{\partial u_m}{\partial x} \right). \quad (23.5)$$

If k and C are treated as constants, Eq. (23.2) reduces to

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \frac{k}{C} \frac{\partial^2 u_m}{\partial x^2} - \frac{\partial q_b}{\partial x}. \quad (23.6)$$

This equation has two conceptually different right-hand-side terms:

$$\frac{k}{C} \frac{\partial^2 u_m}{\partial x^2} \quad \text{is local diffusion of the scattered carrier energy,} \quad (23.7)$$

$$-\frac{\partial q_b}{\partial x} \quad \text{is local gain or loss caused by the divergence of ballistic flux.} \quad (23.8)$$

The ballistic term is not an optional correction. It is the mechanism by which boundary-emitted carriers deposit energy into the interior after streaming and scattering.

23.3 Initial and boundary data in physical variables

At $t = 0$, the film is at equilibrium temperature T_0 . In a constant- C model, the reference total internal energy is

$$u_0 = CT_0, \quad (23.9)$$

up to an arbitrary additive reference. The physically relevant perturbation is measured relative to this state. The initial conditions are

$$u(x, 0) = u_0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0. \quad (23.10)$$

Because the model separates total energy into ballistic and medium pieces, these conditions are implemented by subtracting equilibrium contributions:

$$u_m(x, 0) - u_{m0}(x) = 0, \quad u_b(x, 0) - u_{b0}(x) = 0. \quad (23.11)$$

The left boundary condition is not “ $T(0, t) = T_1$ ” in the Fourier sense. Instead, the left boundary emits a population of carriers characterized by T_1 . The right boundary emits a population characterized by T_0 . The medium component satisfies the Marshak-type boundary condition derived from the one-sided incoming angular population:

$$\left(\tau \frac{\partial u_m}{\partial t} + u_m \right)_{x=0} = -\frac{\Lambda}{3} \left. \frac{\partial u_m}{\partial x} \right|_{x=0}, \quad (23.12)$$

$$\left(\tau \frac{\partial u_m}{\partial t} + u_m \right)_{x=L} = +\frac{\Lambda}{3} \left. \frac{\partial u_m}{\partial x} \right|_{x=L}, \quad (23.13)$$

where the sign changes because the outward normal is $-\hat{x}$ at $x = 0$ and $+\hat{x}$ at $x = L$. Some papers write these signs using a local outward normal convention. The safest implementation rule is to evaluate the normal derivative $\nabla u_m \cdot \mathbf{n}$ at each boundary.

Physical meaning of the boundary condition. The boundary condition does not say that the medium energy is fixed at the wall. It says that the diffusive medium flux at a boundary is made only of the carriers incident on that boundary from inside the domain. Carriers emitted by the wall are assigned to the ballistic component.

23.4 Nondimensional variables

The thin-film equation becomes much more interpretable after nondimensionalization. Define

$$\eta = \frac{x}{L}, \quad t^* = \frac{t}{\tau}, \quad \text{Kn} = \frac{\Lambda}{L}, \quad \Delta T = T_1 - T_0. \quad (23.14)$$

Here η measures position as a fraction of the film thickness, t^* measures time in units of one scattering time, and Kn measures how far a carrier travels before scattering compared with the film thickness.

The normalized medium energy is

$$\theta_m(\eta, t^*) = \frac{u_m(x, t) - u_{m0}(x)}{C\Delta T}. \quad (23.15)$$

The normalized ballistic energy and total energy are

$$\theta_b(\eta, t^*) = \frac{u_b(x, t) - u_{b0}(x)}{C\Delta T}, \quad \theta(\eta, t^*) = \theta_m(\eta, t^*) + \theta_b(\eta, t^*). \quad (23.16)$$

The normalized ballistic flux is

$$q_b^*(\eta, t^*) = \frac{q_b(x, t) - q_{b0}(x)}{Cv\Delta T}. \quad (23.17)$$

The derivative conversions are

$$\frac{\partial}{\partial t} = \frac{1}{\tau} \frac{\partial}{\partial t^*}, \quad \frac{\partial^2}{\partial t^2} = \frac{1}{\tau^2} \frac{\partial^2}{\partial t^{*2}}, \quad (23.18)$$

$$\frac{\partial}{\partial x} = \frac{1}{L} \frac{\partial}{\partial \eta}, \quad \frac{\partial^2}{\partial x^2} = \frac{1}{L^2} \frac{\partial^2}{\partial \eta^2}. \quad (23.19)$$

We also use the kinetic-theory estimate for thermal conductivity:

$$k \approx \frac{1}{3} C v \Lambda. \quad (23.20)$$

The factor $1/3$ appears because in an isotropic three-dimensional carrier gas, only one third of the squared velocity contributes to transport along a chosen coordinate direction:

$$\langle v_x^2 \rangle = \frac{1}{3} v^2. \quad (23.21)$$

Since $\Lambda = v\tau$, Eq. (23.20) is equivalently

$$\frac{k}{C} = \frac{v\Lambda}{3} = \frac{v^2\tau}{3}. \quad (23.22)$$

23.5 Step-by-step nondimensionalization of the governing equation

Start with

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \frac{k}{C} \frac{\partial^2 u_m}{\partial x^2} - \frac{\partial q_b}{\partial x}. \quad (23.23)$$

Using $u_m - u_{m0} = C\Delta T\theta_m$, the left-hand side becomes

$$\begin{aligned}\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} &= \tau C\Delta T \frac{1}{\tau^2} \frac{\partial^2 \theta_m}{\partial t^{*2}} + C\Delta T \frac{1}{\tau} \frac{\partial \theta_m}{\partial t^*} \\ &= \frac{C\Delta T}{\tau} \left(\frac{\partial^2 \theta_m}{\partial t^{*2}} + \frac{\partial \theta_m}{\partial t^*} \right).\end{aligned}\quad (23.24)$$

The diffusive term becomes

$$\frac{k}{C} \frac{\partial^2 u_m}{\partial x^2} = \frac{k}{C} C\Delta T \frac{1}{L^2} \frac{\partial^2 \theta_m}{\partial \eta^2} \quad (23.25)$$

$$= k\Delta T \frac{1}{L^2} \frac{\partial^2 \theta_m}{\partial \eta^2}. \quad (23.26)$$

Divide this by the common left-hand-side scale $C\Delta T/\tau$:

$$\frac{k\Delta T/L^2}{C\Delta T/\tau} = \frac{k\tau}{CL^2} = \frac{1}{L^2} \left(\frac{v\Lambda}{3} \right) \tau = \frac{v^2\tau^2}{3L^2} = \frac{\Lambda^2}{3L^2} = \frac{\text{Kn}^2}{3}. \quad (23.27)$$

Thus the nondimensional diffusion coefficient is $\text{Kn}^2/3$.

For the ballistic term, use $q_b - q_{b0} = Cv\Delta T q_b^*$:

$$\frac{\partial q_b}{\partial x} = Cv\Delta T \frac{1}{L} \frac{\partial q_b^*}{\partial \eta}. \quad (23.28)$$

Divide by $C\Delta T/\tau$:

$$\frac{Cv\Delta T/L}{C\Delta T/\tau} = \frac{v\tau}{L} = \frac{\Lambda}{L} = \text{Kn}. \quad (23.29)$$

Therefore Eq. (23.6) becomes

$$\boxed{\frac{\partial^2 \theta_m}{\partial t^{*2}} + \frac{\partial \theta_m}{\partial t^*} = \frac{\text{Kn}^2}{3} \frac{\partial^2 \theta_m}{\partial \eta^2} - \text{Kn} \frac{\partial q_b^*}{\partial \eta}}. \quad (23.30)$$

Slide takeaway. The nondimensional thin-film equation exposes the controlling parameter. The same physical equation changes character as $\text{Kn} = \Lambda/L$ changes. This is why Module 4 validation must sweep Kn .

23.6 Dimensionless initial and boundary conditions

The normalized initial conditions are

$$\theta(\eta, 0) = 0, \quad \left. \frac{\partial \theta}{\partial t^*} \right|_{t^*=0} = 0. \quad (23.31)$$

A practical implementation usually initializes the nonequilibrium medium and ballistic perturbations as

$$\theta_m(\eta, 0) = 0, \quad \theta_b(\eta, 0) = 0, \quad (23.32)$$

with the boundary step entering through the emitted ballistic intensity.

Using $x = L\eta$, $\Lambda/L = \text{Kn}$, and $u_m - u_{m0} = C\Delta T\theta_m$, the Marshak-type boundary conditions become

$$\left(\frac{\partial \theta_m}{\partial t^*} + \theta_m \right)_{\eta=0} = -\frac{\text{Kn}}{3} \frac{\partial \theta_m}{\partial \eta} \Big|_{\eta=0}, \quad (23.33)$$

$$\left(\frac{\partial \theta_m}{\partial t^*} + \theta_m \right)_{\eta=1} = +\frac{\text{Kn}}{3} \frac{\partial \theta_m}{\partial \eta} \Big|_{\eta=1}. \quad (23.34)$$

Again, the sign convention is most safely implemented through the outward normal derivative.

23.7 Ballistic intensity in one dimension

Let $\mu = \cos \theta$ be the directional cosine relative to the positive x -direction. A carrier emitted from the left boundary and traveling with $0 < \mu < 1$ reaches location x after flight time

$$t_{\text{flight}} = \frac{x}{\mu v}. \quad (23.35)$$

The path length is x/μ , not x , because a shallow-angle carrier travels diagonally through the film. The probability-like attenuation factor for no scattering along that path is

$$\exp\left(-\frac{x/\mu}{\Lambda}\right) = \exp\left(-\frac{x}{\mu\Lambda}\right). \quad (23.36)$$

Therefore the left-emitted ballistic intensity has the form

$$I_b^{(+)}(x, t, \mu) = I_{b1} \left(t - \frac{x}{\mu v} \right) \exp\left(-\frac{x}{\mu\Lambda}\right), \quad 0 < \mu < 1. \quad (23.37)$$

Similarly, for right-emitted carriers moving left, $-1 < \mu < 0$,

$$I_b^{(-)}(x, t, \mu) = I_{b2} \left(t - \frac{L-x}{|\mu|v} \right) \exp\left(-\frac{L-x}{|\mu|\Lambda}\right), \quad -1 < \mu < 0. \quad (23.38)$$

The time arguments are retarded boundary times. They say that the interior point sees what the boundary emitted earlier, not what the boundary emits at the current time.

23.8 Finite arrival time and the lower angular limit

For a step heating applied at the left boundary at $t = 0$, a left-emitted carrier contributes at position x only if

$$t - \frac{x}{\mu v} \geq 0. \quad (23.39)$$

This gives

$$\mu \geq \mu_t, \quad \mu_t = \frac{x}{vt}. \quad (23.40)$$

In nondimensional variables, because $x = L\eta$, $t = \tau t^*$, and $v\tau = \Lambda = \text{Kn}L$,

$$\boxed{\mu_t = \frac{\eta}{\text{Kn} t^*}}. \quad (23.41)$$

This expression has a direct physical interpretation:

- if $\mu_t > 1$, no carrier direction has reached η yet;
- if $0 < \mu_t < 1$, only sufficiently forward-directed carriers have arrived;
- if $\mu_t \leq 0$, all forward directions are allowed for a sustained step source.

23.9 Ballistic flux contribution for the step-heated film

The heat flux is the directional energy flow. In one dimension, each carrier direction contributes a factor μ because only the x -component of velocity transports energy across a plane normal to x . For the left-emitted perturbation, the normalized ballistic flux contribution is

$$\boxed{q_b^*(\eta, t^*) = \frac{1}{2} \int_{\mu_t}^1 \mu \exp\left(-\frac{\eta}{\mu \text{Kn}}\right) d\mu}, \quad 0 < \mu_t < 1. \quad (23.42)$$

The factor $1/2$ comes from angular averaging over a half-space in the one-dimensional slab reduction. When $\mu_t > 1$, the integral is zero because no ballistic carrier from the heated boundary has arrived. When $\mu_t < 0$, the lower limit is effectively zero for the forward half-space.

The derivative that enters the medium-energy equation is

$$\frac{\partial q_b^*}{\partial \eta}. \quad (23.43)$$

This term is negative where the forward ballistic flux decays with distance. Because the PDE contains $-\text{Kn} \partial q_b^* / \partial \eta$, attenuation of ballistic flux becomes a positive source for the scattered medium energy. In words: ballistic carriers lose energy from the unscattered population by scattering, and that energy appears in the medium population.

23.10 Ballistic internal energy contribution

The ballistic internal energy contribution is similar to the flux contribution but without the directional projection factor μ . A representative normalized left-boundary contribution has the form

$$\theta_b(\eta, t^*) = \frac{1}{2} \int_{\mu_t}^1 \exp\left(-\frac{\eta}{\mu K_n}\right) d\mu, \quad 0 < \mu_t < 1. \quad (23.44)$$

The total normalized energy-temperature response is then

$$\theta(\eta, t^*) = \theta_m(\eta, t^*) + \theta_b(\eta, t^*). \quad (23.45)$$

This is the quantity most analogous to the temperature plotted in thin-film comparisons, with the warning that it is an equivalent internal-energy temperature in the nonequilibrium regime.

23.11 Equilibrium subtraction and why it is not cosmetic

The P1 approximation is an approximation to the angular distribution of the medium component. Because of this approximation, the separate equilibrium pieces u_{b0} and u_{m0} do not necessarily sum to a perfectly uniform equilibrium energy if handled naively. Chen's normalization procedure subtracts the reference equilibrium ballistic and medium components:

$$\theta_m = \frac{u_m - u_{m0}}{C\Delta T}, \quad (23.46)$$

$$\theta_b = \frac{u_b - u_{b0}}{C\Delta T}, \quad (23.47)$$

$$q_b^* = \frac{q_b - q_{b0}}{Cv\Delta T}. \quad (23.48)$$

This focuses the calculation on the nonequilibrium perturbation caused by the boundary heating and reduces inconsistency associated with the approximate angular closure.

One way to see the issue is to imagine the whole film at uniform equilibrium temperature T_0 . Physically, the total energy should be uniform. However, after splitting the angular distribution into a boundary-originating piece and an approximate P1 medium piece, the individual reconstructed pieces may have spatial dependence near boundaries. The subtraction removes the reference spatial structure and leaves only the perturbation that matters for the transient experiment.

Physical meaning. The equilibrium subtraction is not a mathematical trick to hide an error. It is a bookkeeping correction that keeps the approximate split model focused on the nonequilibrium change from the reference state.

23.12 What should be plotted in the Module 4 validation

For $\text{Kn} = 0.01, 0.1, 1, 10$, the validation should plot:

1. $\theta_m(\eta, t^*)$: scattered medium energy,
2. $\theta_b(\eta, t^*)$: ballistic energy contribution,
3. $\theta(\eta, t^*) = \theta_m + \theta_b$: total equivalent temperature response,
4. $q_b^*(\eta, t^*)$: ballistic heat flux,
5. total heat flux history at the boundaries,
6. comparison against Fourier and Cattaneo predictions.

The expected trend is:

$$\text{Kn} \ll 1 : \quad \text{Fourier-like behavior after very short transients,} \quad (23.49)$$

$$\text{Kn} \sim 1 : \quad \text{strong ballistic-diffusive correction,} \quad (23.50)$$

$$\text{Kn} \gg 1 : \quad \text{dominant boundary-originating ballistic transport.} \quad (23.51)$$

24 Dedicated comparison: why Fourier and Cattaneo fail

24.1 Failure mode of Fourier conduction

Fourier conduction fails in the quasi-ballistic regime for three related reasons:

1. **Infinite speed.** The parabolic heat equation predicts an instantaneous response everywhere.
2. **No carrier flight time.** It cannot represent the delay $(s - s_0)/v$ between boundary emission and interior arrival.
3. **No nonlocal boundary memory.** The flux is controlled only by ∇T at the same point, not by the boundary state from which unscattered carriers originated.

As $t \rightarrow 0^+$, Fourier theory can predict unrealistically large surface heat flux because it assumes an immediately established temperature gradient.

24.2 Failure mode of Cattaneo conduction

Cattaneo conduction improves Fourier theory by adding a heat-flux relaxation time, but it still fails to represent the population split. Its limitations are:

1. **All flux is treated as one local variable.** There is no separate $\mathbf{q}_b$ that remembers boundary origin.
2. **Ballistic transport is replaced by a thermal wave.** The resulting wave speed is finite, but a thermal wave is not the same as angularly resolved particles arriving at retarded times.
3. **Boundary conditions remain problematic.** A prescribed local temperature at a boundary is not necessarily the same as a boundary-emitted carrier distribution.
4. **Artificial oscillations can occur.** Hyperbolic heat equations can generate nonphysical flux oscillations in regimes where the Boltzmann solution shows smooth ballistic arrival and scattering.

24.3 Why the ballistic-diffusive model is better

The ballistic-diffusive model keeps the strongest advantages of both descriptions:

Boltzmann-like feature: retarded boundary-originating ballistic flux $\mathbf{q}_b$, (24.1)

Diffusion-like feature: ordinary space-time PDE for the scattered medium energy u_m . (24.2)

It avoids a full phase-space solve while retaining the physical mechanism that Fourier and Cattaneo miss: unscattered carriers crossing a finite structure.

25 Weak form for FEM implementation

The FEM starts from Eq. (19.9). For compactness, define

$$\alpha = \frac{k}{C}, \quad S_{\text{BD}} = -\nabla \cdot \mathbf{q}_b + \dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t}. \quad (25.1)$$

Then

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot (\alpha \nabla u_m) + S_{\text{BD}}. \quad (25.2)$$

Move everything except the source to the left:

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} - \nabla \cdot (\alpha \nabla u_m) = S_{\text{BD}}. \quad (25.3)$$

Multiply by a test function w and integrate over Ω :

$$\int_{\Omega} w \tau \frac{\partial^2 u_m}{\partial t^2} dV + \int_{\Omega} w \frac{\partial u_m}{\partial t} dV - \int_{\Omega} w \nabla \cdot (\alpha \nabla u_m) dV = \int_{\Omega} w S_{\text{BD}} dV. \quad (25.4)$$

Integrate the diffusion term by parts:

$$-\int_{\Omega} w \nabla \cdot (\alpha \nabla u_m) dV = \int_{\Omega} \nabla w \cdot \alpha \nabla u_m dV - \int_{\partial\Omega} w \alpha \nabla u_m \cdot \mathbf{n} dA. \quad (25.5)$$

Thus the weak form is

$$\boxed{\int_{\Omega} w \tau \frac{\partial^2 u_m}{\partial t^2} dV + \int_{\Omega} w \frac{\partial u_m}{\partial t} dV + \int_{\Omega} \nabla w \cdot \alpha \nabla u_m dV = \int_{\Omega} w S_{BD} dV + \int_{\partial\Omega} w \alpha \nabla u_m \cdot \mathbf{n} dA.} \quad (25.6)$$

Boundary conditions determine how the boundary integral is evaluated.

26 Matrix form

Approximate the unknown by basis functions:

$$u_m(\mathbf{r}, t) \approx \sum_{j=1}^N U_j(t) N_j(\mathbf{r}). \quad (26.1)$$

Choose test functions $w = N_i$. The matrix equation becomes

$$\boxed{\tau M \ddot{\mathbf{U}} + M \dot{\mathbf{U}} + K \mathbf{U} = \mathbf{F}_{BD} + \mathbf{F}_{\partial\Omega}.} \quad (26.2)$$

The matrices are

$$M_{ij} = \int_{\Omega} N_i N_j dV, \quad (26.3)$$

$$K_{ij} = \int_{\Omega} \nabla N_i \cdot \alpha \nabla N_j dV, \quad (26.4)$$

$$(F_{BD})_i = \int_{\Omega} N_i S_{BD} dV, \quad (26.5)$$

$$(F_{\partial\Omega})_i = \int_{\partial\Omega} N_i \alpha \nabla u_m \cdot \mathbf{n} dA. \quad (26.6)$$

This second-order-in-time system can be solved directly or rewritten as a first-order system by defining $\mathbf{V} = \dot{\mathbf{U}}$:

$$\dot{\mathbf{U}} = \mathbf{V}, \quad (26.7)$$

$$\tau M \dot{\mathbf{V}} = \mathbf{F}_{BD} + \mathbf{F}_{\partial\Omega} - M \mathbf{V} - K \mathbf{U}. \quad (26.8)$$

27 Module 4 FEM specialization for the MATLAB implementation

The previous weak and matrix forms describe the general finite-element structure. The MATLAB implementation added for Module 4 uses a concrete specialization: a two-dimensional rectangular

silicon cross section, divided into three-node triangular elements, with nodal temperature as the primary unknown. This section is included so that the implemented code can be explained from first principles rather than described only as a numerical black box.

27.1 Strong form solved by the FEM path

The FEM path uses the temperature version of the reduced ballistic-diffusive equation,

$$\tau C \frac{\partial^2 T}{\partial t^2} + C \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) = Q + \tau \frac{\partial Q}{\partial t} - \nabla \cdot \mathbf{q}_b, \quad (27.1)$$

where T is the medium-equivalent temperature, C is volumetric heat capacity, k is thermal conductivity, Q is volumetric heat generation, and $\mathbf{q}_b$ is the reduced ballistic heat-flux contribution. The right-hand side is grouped as

$$S_T = Q + \tau \frac{\partial Q}{\partial t} - \nabla \cdot \mathbf{q}_b. \quad (27.2)$$

Therefore Eq. (27.1) becomes

$$\tau C T_{tt} + C T_t - \nabla \cdot (k \nabla T) = S_T. \quad (27.3)$$

This form is the direct thermal analogue of the FEM equations used in Modules 2 and 3: the PDE is converted into a weak statement, then into element matrices, then assembled into a global matrix problem.

27.2 Triangular mesh and nodal temperature unknowns

The two-dimensional domain $\Omega = [0, L_x] \times [0, L_y]$ is discretized into a structured triangular mesh. A rectangular grid of nodes is first generated, and each rectangular cell is split into two counter-clockwise triangles. The mesh data are

$$\text{nodes} = \{(x_i, y_i)\}_{i=1}^{N_n}, \quad (27.4)$$

$$\text{elements} = \{(i_1, i_2, i_3)_e\}_{e=1}^{N_e}, \quad (27.5)$$

where N_n is the number of nodes and N_e is the number of triangular elements.

The FEM approximation writes the temperature as

$$T_h(x, y, t) = \sum_{j=1}^{N_n} N_j(x, y) T_j(t), \quad (27.6)$$

where N_j is the global shape function associated with node j , and $T_j(t)$ is the nodal temperature.

Inside one three-node triangle, the local form is

$$T_h^{(e)}(x, y, t) = N_1^{(e)}(x, y)T_1^{(e)}(t) + N_2^{(e)}(x, y)T_2^{(e)}(t) + N_3^{(e)}(x, y)T_3^{(e)}(t). \quad (27.7)$$

For a linear triangle,

$$N_a^{(e)}(x, y) = a_a + b_a x + c_a y, \quad a = 1, 2, 3, \quad (27.8)$$

and the gradient $\nabla N_a^{(e)}$ is constant inside the element.

27.3 Weak form in temperature variables

Multiply Eq. (27.3) by a test function w and integrate over the domain:

$$\int_{\Omega} w \tau C T_{tt} \, dA + \int_{\Omega} w C T_t \, dA - \int_{\Omega} w \nabla \cdot (k \nabla T) \, dA = \int_{\Omega} w S_T \, dA. \quad (27.9)$$

Integrating the conduction term by parts gives

$$\boxed{\int_{\Omega} w \tau C T_{tt} \, dA + \int_{\Omega} w C T_t \, dA + \int_{\Omega} \nabla w \cdot k \nabla T \, dA = \int_{\Omega} w S_T \, dA + \int_{\partial\Omega} w k \nabla T \cdot \mathbf{n} \, ds.} \quad (27.10)$$

For the first MATLAB implementation, the default boundary condition is homogeneous natural conductive flux,

$$k \nabla T \cdot \mathbf{n} = 0 \quad \text{on insulated or symmetry boundaries,} \quad (27.11)$$

so the boundary term vanishes unless an explicit Neumann thermal flux is added later. Fixed-temperature boundaries can also be imposed strongly by replacing the corresponding matrix rows with prescribed nodal values.

27.4 Element heat-capacity matrix

Choosing $w = N_i$ and substituting Eq. (27.6) gives the heat-capacity matrix

$$(M_C)_{ij} = \int_{\Omega} C N_i N_j \, dA. \quad (27.12)$$

For one linear triangular element with constant C ,

$$\boxed{M_C^{(e)} = \frac{C A_e}{12} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.} \quad (27.13)$$

This is the thermal analogue of a consistent mass matrix. It maps nodal temperature and nodal temperature rate into element thermal energy storage.

27.5 Element conductivity matrix

The conduction matrix is

$$(K_T)_{ij} = \int_{\Omega} k \nabla N_i \cdot \nabla N_j \, dA. \quad (27.14)$$

For a linear triangle with constant k , the gradients are constant, so

$$\boxed{K_T^{(e)} = k A_e B_e B_e^T}, \quad (27.15)$$

where

$$B_e = \begin{bmatrix} \nabla N_1^{(e)} \\ \nabla N_2^{(e)} \\ \nabla N_3^{(e)} \end{bmatrix}. \quad (27.16)$$

The physical meaning is direct: stronger thermal conductivity k or sharper basis-function gradients increase the energetic penalty for strong temperature curvature.

27.6 Element source vector

The right-hand-side vector is

$$(F_T)_i = \int_{\Omega} N_i S_T \, dA. \quad (27.17)$$

At the element level,

$$\mathbf{F}_T^{(e)} = \int_{\Omega_e} \mathbf{N}^T \left(Q + \tau \frac{\partial Q}{\partial t} - \nabla \cdot \mathbf{q}_b \right) dA. \quad (27.18)$$

The implemented reduced solver evaluates S_T at nodes and uses the same consistent triangular mass weighting used for scalar source projection. This is the correct first implementation because it treats the thermal source as a distributed load rather than injecting heat at one grid point.

27.7 Global semi-discrete FEM equation

After element assembly, the global system is

$$\boxed{\tau M_C \ddot{\mathbf{T}} + M_C \dot{\mathbf{T}} + K_T \mathbf{T} = \mathbf{F}_T(t)}. \quad (27.19)$$

Here $\mathbf{T}$ is the vector of nodal temperatures. This equation has the same assembly logic as the Module 2 Poisson solver and the Module 3 diffusion-reaction solver, but it has a second time derivative because the ballistic-diffusive reduction retains a relaxation-time correction.

27.8 Backward-Euler time stepping used in MATLAB

Define the nodal temperature-rate vector

$$\mathbf{V} = \dot{\mathbf{T}}. \quad (27.20)$$

The first-order system is

$$\dot{\mathbf{T}} = \mathbf{V}, \quad (27.21)$$

$$\tau M_C \dot{\mathbf{V}} + M_C \mathbf{V} + K_T \mathbf{T} = \mathbf{F}_T(t). \quad (27.22)$$

Using backward Euler,

$$\mathbf{V}^{n+1} = \frac{\mathbf{T}^{n+1} - \mathbf{T}^n}{\Delta t}, \quad (27.23)$$

and substituting into the second equation gives the linear solve

$$\left(\frac{\tau}{\Delta t^2} M_C + \frac{1}{\Delta t} M_C + K_T \right) \mathbf{T}^{n+1} = \mathbf{F}_T^{n+1} + \left(\frac{\tau}{\Delta t^2} + \frac{1}{\Delta t} \right) M_C \mathbf{T}^n + \frac{\tau}{\Delta t} M_C \mathbf{V}^n. \quad (27.24)$$

Then

$$\mathbf{V}^{n+1} = \frac{\mathbf{T}^{n+1} - \mathbf{T}^n}{\Delta t}. \quad (27.25)$$

In the diffusive limit $\tau \rightarrow 0$, Eq. (27.24) reduces to the ordinary implicit heat-equation step,

$$\left(\frac{1}{\Delta t} M_C + K_T \right) \mathbf{T}^{n+1} = \mathbf{F}_T^{n+1} + \frac{1}{\Delta t} M_C \mathbf{T}^n. \quad (27.26)$$

This limiting behavior is an important verification check: the FEM ballistic-diffusive solver must collapse to a standard FEM heat solver when the relaxation-time and ballistic terms are removed.

27.9 How the ballistic term is represented in the first FEM path

A full ballistic treatment would require angular quadrature or ray tracing. The first executable Module 4 FEM path instead uses the same reduced rectangular-domain closure as the structured-grid Module 4 solver: each boundary can emit an attenuated, retarded heat-carrier contribution traveling approximately normal to that boundary. The code evaluates $\mathbf{q}_b$ at mesh nodes and approximates $-\nabla \cdot \mathbf{q}_b$ as a nodal source term.

This is not the final microscopic ballistic model. It is a deliberately reduced closure that gives the FEM system the correct place to include finite-flight-time and boundary-memory effects while keeping the first solver simple enough to verify.

27.10 Verification tests for the FEM path

The FEM-specific tests are:

1. **Uniform equilibrium:** with uniform initial temperature, no source, no ballistic emission, and natural no-flux boundaries, the solution should remain uniform.
2. **Gaussian diffusion in the Fourier limit:** setting $\tau = 0$ and $\mathbf{q}_b = 0$ should produce ordinary thermal smoothing while conserving total thermal energy under insulated boundaries.
3. **Uniform volumetric source:** a spatially uniform source should keep the field spatially uniform and should match the analytic uniform response in time.
4. **Boundary-emission demonstration:** a boundary-emitted ballistic pulse should enter the FEM system through $-\nabla \cdot \mathbf{q}_b$ and generate a localized thermal response.

These checks make the FEM implementation defensible: they test storage, conduction, source projection, boundary handling, and the diffusive-limit reduction separately.

27.11 MATLAB implementation map

The Module 4 FEM implementation corresponds to the derivation as follows:

FEM concept	MATLAB implementation target
Triangular mesh	<code>make_rectangular_tri_mesh_2d.m</code>
Element heat-capacity matrix	<code>compute_element_heat_capacity_triangle.m</code>
Element conductivity matrix	<code>compute_element_conductivity_triangle.m</code>
Global matrix assembly	<code>assemble_thermal_fem_2d.m</code>
Source projection	<code>assemble_thermal_source_fem_2d.m</code> and <code>evaluate_module4_fem_source_2d.m</code>
Implicit time stepping	<code>solve_ballistic_diffusive_thermal_fem_2d.m</code>
Top-level driver	<code>main_module4_fem_ballistic_diffusive_thermal.m</code>
Verification tests	<code>test_module4_fem_uniform_equilibrium_2d.m</code> , <code>test_module4_fem_gaussian_diffusion_2d.m</code> , and <code>test_module4_fem_uniform_source_2d.m</code>

28 Practical implementation strategy for this project

The Module 4 implementation should be built in layers rather than jumping directly to a realistic chip geometry. The defensible sequence is:

1. verify the linear-triangle FEM heat-capacity and conductivity matrices on simple rectangular domains,
2. verify the diffusive limit by setting $\tau = 0$ and $\mathbf{q}_b = 0$,
3. verify the ballistic-diffusive relaxation term with spatially uniform sources where analytic solutions exist,
4. add the reduced boundary-emission ballistic source $-\nabla \cdot \mathbf{q}_b$,
5. compare the FEM path against the existing structured-grid Module 4 path and the archived Fourier baseline,
6. only then couple the temperature field back into Module 3 kinetic coefficients and later Module 5 transport coefficients.

The primary FEM code target is therefore

$$\text{main_module4_fem_ballistic_diffusive_thermal.m.} \quad (28.1)$$

This driver should generate nodal temperature fields, source fields, ballistic-divergence fields, scalar history metrics, and verification summaries. The FEM path is not meant to replace the structured-grid path immediately. It provides an interview-defensible and geometry-extensible formulation that can later be generalized to irregular device domains.

29 How this connects to the superconducting-qubit project

The final goal of this project is not simply to solve a heat equation. The larger goal is to understand how geometry and materials influence energy flow and radiation-induced degradation in silicon device structures. Module 4 supplies the thermal transport field needed by the later defect, electrostatic, and carrier-transport modules.

In a qubit device, local energy can be deposited near a Josephson junction, in a capacitor pad, in a substrate, or at an interface. The early-time flow of that energy can determine whether high-energy phonons or quasiparticles reach the sensitive junction region. A trap or heat sink changes the geometry and boundary conditions of the transport problem. Therefore, a model that captures finite carrier flight, boundary memory, and diffusion after scattering is more appropriate than immediately assuming ordinary Fourier conduction everywhere.

30 Assumptions and limitations

The ballistic-diffusive model used here is still an approximation. Its main assumptions are:

1. **Particle-like carriers.** The model assumes a particle transport picture. It does not describe coherent wave interference of phonons.
2. **Relaxation-time approximation.** Detailed scattering physics is compressed into τ_ω or an averaged τ .
3. **Angular P1 closure for scattered carriers.** The medium component is approximated as nearly isotropic plus first-order anisotropy.
4. **Averaged material properties.** The first implementation uses effective C , k , v , τ , and Λ , rather than fully mode-resolved phonon bands.
5. **Boundary modeling matters.** The ballistic source depends strongly on emitted/reflected boundary intensity. A poor boundary model can dominate the result.
6. **Room-temperature foundation.** Cryogenic superconducting changes, quasiparticle generation, recombination, and electron-phonon nonequilibrium are deferred to later modules.

31 Summary of the derivation

The derivation can be summarized as follows:

1. Start from local energy conservation: $\partial u / \partial t + \nabla \cdot \mathbf{q} = \dot{q}_e$.
2. Describe heat carriers using the BTE under RTA: $\partial f / \partial t + \mathbf{v} \cdot \nabla f = -(f - f_0) / \tau$.
3. Convert to intensity notation so that angular moments produce u and $\mathbf{q}$.
4. Split the intensity into $I_\omega = I_{b\omega} + I_{m\omega}$.
5. Treat $I_{b\omega}$ by characteristics, giving retarded boundary emission and exponential attenuation.
6. Treat $I_{m\omega}$ using the P1 diffusion approximation.
7. Derive a Cattaneo-like relation for only the medium flux: $\tau \partial \mathbf{q}_m / \partial t + \mathbf{q}_m = -(k/C) \nabla u_m$.
8. Combine with energy conservation to obtain

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \nabla \cdot \left(\frac{k}{C} \nabla u_m \right) - \nabla \cdot \mathbf{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right).$$

9. Convert this equation into a FEM weak form for numerical implementation.

32 Lecture-ready slide translation for Module 4

This section converts the derivation into presentation logic. Each item corresponds to one possible slide or slide cluster.

Slide 1: Why ordinary heat conduction breaks down. Physical question: When does $\mathbf{q} = -k\nabla T$ stop being enough? Governing idea: Fourier conduction assumes local equilibrium and local flux response. Approximation being challenged: heat flux is determined only by the local temperature gradient. Figure to show: a chip-scale length L compared with carrier mean free path Λ .

Slide 2: Heat carriers and mean free path. Physical question: What actually carries thermal energy? Governing idea: phonons or other carriers move with group velocity v and scatter after time τ . Key equation: $\Lambda = v\tau$. Figure to show: carrier trajectories with short and long mean free paths.

Slide 3: The distribution function. Physical question: What replaces temperature when the system is not locally equilibrated? Governing idea: use $f(t, \mathbf{r}, \mathbf{k})$, a phase-space occupation. Approximation: local equilibrium is represented by f_0 , but the actual distribution may be far from f_0 . Figure to show: isotropic equilibrium distribution versus directional nonequilibrium distribution.

Slide 4: Boltzmann transport equation. Physical question: How does the microscopic carrier distribution evolve? Governing equation: $\partial f / \partial t + \mathbf{v} \cdot \nabla f = -(f - f_0) / \tau$. Approximation: relaxation-time approximation. Figure to show: streaming plus scattering toward equilibrium.

Slide 5: Fourier as a limiting model. Physical question: How does Fourier conduction emerge? Governing idea: when $\Lambda \ll L$, many scattering events erase directional memory, and the heat flux becomes local. Approximation: first-order near-equilibrium response to ∇T . Figure to show: many scattering events over one device length.

Slide 6: Why Cattaneo is not enough. Physical question: Does finite flux relaxation solve the nanoscale problem? Governing equation: $\tau \partial \mathbf{q} / \partial t + \mathbf{q} = -k \nabla T$. Key limitation: it gives finite propagation speed but still lacks boundary-originating ballistic carriers. Figure to show: thermal wave picture versus particle flight picture.

Slide 7: Ballistic-diffusive split. Physical question: How can we keep ballistic memory without solving full phase space? Governing split: $I_\omega = I_{b\omega} + I_{m\omega}$. Approximation: treat boundary-originating carriers by characteristics and scattered carriers by diffusion/P1 closure. Figure to show: carriers from wall versus carriers scattered inside the domain.

Slide 8: Final ballistic-diffusive PDE. Physical question: What equation will the FEM solve? Governing equation: medium energy obeys a Cattaneo-like PDE forced by $-\nabla \cdot \mathbf{q}_b$. Key message: the ballistic term is the physical memory missing from Fourier and Cattaneo models. Figure to show: PDE box with u_m , plus ray-based computation of $\mathbf{q}_b$.

Slide 9: Thin-film benchmark. Physical question: How do we validate the implementation? Governing parameter: $\text{Kn} = \Lambda/L$. Approximation being tested: reduced ballistic-diffusive equation versus Fourier and Cattaneo limits. Figure to show: profiles for $\text{Kn} = 0.01, 0.1, 1, 10$.

Slide 10: Relevance to superconducting qubits. Physical question: Why does this matter for qubit poisoning? Governing idea: energy deposited in a small region can move through phonons and other carriers before local equilibrium is established. Design relevance: heat sinks and traps should be evaluated by how they alter energy and quasiparticle transport near the junction-sensitive region.

33 Module 4 implementation checklist

Before coupling Module 4 to Modules 3, 5, and 6, the project should satisfy the following Module 4 checks:

1. The code can compute $q_b^*(\eta, t^*)$ with the correct finite-arrival lower limit $\mu_t = \eta/(\text{Kn}t^*)$.
2. The code can solve Eq. (23.30) for several Kn values.
3. The diffusive limit trends toward Fourier-like behavior for $\text{Kn} \ll 1$ after early transients.
4. The quasi-ballistic cases show delayed energy arrival and boundary-memory effects.
5. The Cattaneo comparison is included to show why finite propagation speed alone is not the full physics.
6. The plotted quantity is clearly labeled as equivalent internal-energy temperature, not blindly treated as local thermodynamic temperature.

34 Module 4 interpretation for RadiationEffects_proj2

The purpose of Module 4 in this project is different from the original thin-film validation use case. Here the ballistic-diffusive framework is used as a reduced thermal model for a damaged silicon device cross section. The thermal field is not an isolated curiosity. It changes defect diffusivities, annealing rates, carrier mobilities, and later the strength of thermal-electrical feedback in the coupled degradation model.

The practical unknown used by the project can be written either as the medium energy density $u_m(x, y, t)$ or as an equivalent temperature field $T_m(x, y, t) = u_m/C$ when a constant volumetric heat capacity is assumed. The ballistic contribution is still treated separately because it carries the boundary-memory and finite-flight-time physics that a local Fourier law cannot represent.

34.1 Why the 2D reduction is the right next step

The existing executable ballistic-diffusive validation model is one-dimensional. That is appropriate for a first thin-film benchmark, but it suppresses lateral heat spreading, hot-spot broadening, and geometry-dependent boundary memory. In RadiationEffects_proj2 the thermal field must live on the same 2D cross-sectional domain used by the electrostatic, defect-evolution, and later carrier-transport modules. Therefore the correct near-term adaptation is not a full angularly resolved 3D solve, but a 2D reduced ballistic-diffusive model on a rectangular silicon domain.

34.2 2D governing equation used as the Module 4 target

With Cartesian coordinates (x, y) , spatially varying thermal properties allowed in principle, and a reduced ballistic source divergence retained explicitly, the medium-energy equation becomes

$$\tau \frac{\partial^2 u_m}{\partial t^2} + \frac{\partial u_m}{\partial t} = \frac{\partial}{\partial x} \left(\frac{k}{C} \frac{\partial u_m}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{k}{C} \frac{\partial u_m}{\partial y} \right) - \nabla \cdot \mathbf{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right). \quad (34.1)$$

If one introduces the equivalent medium temperature $T_m = u_m/C$ and treats C as constant, the same equation can be written as

$$\tau C \frac{\partial^2 T_m}{\partial t^2} + C \frac{\partial T_m}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T_m}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T_m}{\partial y} \right) - \nabla \cdot \mathbf{q}_b + \left(\dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right). \quad (34.2)$$

This is the equation family that should define Module 4 in the revised project plan.

34.3 Source terms in the radiation-effects setting

In the original boundary-heating derivation, much attention naturally falls on hot and cold boundaries. In RadiationEffects_proj2, the source structure is broader. The volumetric source term $\dot{q}_e(x, y, t)$

may represent several distinct physical mechanisms:

1. localized radiation-deposition heating or an effective thermalized fraction of a deposition event,
2. Joule heating from current flow in a later coupled Module 5 or Module 6 solve,
3. optional reaction or defect-reconfiguration heating terms if those are later judged important.

Therefore Module 4 should be implemented with volumetric-source flexibility from the start rather than only with boundary-driven transients.

34.4 How ballistic terms should be interpreted in this project

The ballistic contribution q_b does not mean that the entire silicon device is in a fully ballistic thermal regime. It means the reduced model reserves a separate channel for heat carriers that retain finite-flight-time and boundary-memory effects before joining the scattered medium population. In a large or strongly scattering device, the ballistic term may become weak and the model should reduce smoothly toward a diffusive limit. In a small structure or near a sharp transient, the ballistic term can materially alter early-time fluxes and local temperature rise.

34.5 Coupling to Module 3 defect evolution

Module 3 needs temperature because both defect migration and defect annealing are thermally activated. In the present reduced framework, Module 4 supplies $T(x, y, t)$ to the coefficient maps

$$D = D(T), \quad k_{\text{ann}} = k_{\text{ann}}(T). \quad (34.3)$$

This means a localized thermal transient can accelerate defect motion in one region while leaving another region nearly frozen. The thermal field therefore changes both the rate and the spatial pattern of defect evolution.

34.6 Coupling to Modules 5 and 6

Later modules may use the same temperature field to update carrier mobility, carrier statistics, recombination, and Joule-heating feedback. Module 6 then iterates among defect state, thermal state, electrostatic state, and carrier state until they are mutually consistent. In that coupled setting, Module 4 is no longer only a post-processing thermal step. It becomes one of the active feedback channels of the degradation simulator.

34.7 Verification logic for the revised Module 4

The first 2D implementation should be verified in stages.

1. **Uniform-equilibrium preservation:** a spatially uniform initial state with no source should remain uniform.
2. **Diffusive-limit recovery:** when the relaxation time and ballistic correction are reduced appropriately, the solver should approach the continuum heat-equation result.
3. **Localized-pulse spreading:** a compact source should spread laterally in 2D and exhibit the expected transition from early localized response toward broader thermal diffusion.
4. **Boundary-memory sensitivity:** cases with strong boundary influence should show a measurable difference between the ballistic-diffusive model and a pure Fourier baseline.

These tests are the right bridge between the original 1D benchmark and the intended 2D production module.

34.8 Summary of the revised Module 4 role

In the revised architecture, Module 4 is a single thermal-physics module based on ballistic-diffusive heat conduction rather than a split baseline-versus-comparison pair. The model is still reduced, but it carries a clearer microscopic foundation than a plain continuum heat equation and is therefore a better match to localized and quasi-ballistic thermal transport in silicon device structures. The updated implementation also has a linear triangular FEM path, so the same thermal equation can be explained in terms of mesh generation, shape functions, element heat-capacity matrices, element conductivity matrices, source-vector assembly, and implicit time stepping.

35 References

- [1] G. Chen, “Ballistic-Diffusive Heat-Conduction Equations,” *Physical Review Letters* **86**, 2297–2300 (2001).
- [2] G. Chen, “Ballistic-Diffusive Equations for Transient Heat Conduction From Nano to Macroscales,” *Journal of Heat Transfer* **124**, 320–328 (2002).
- [3] D. D. Joseph and L. Preziosi, “Heat waves,” *Reviews of Modern Physics* **62**, 375–391 (1990).
- [4] A. A. Joshi and A. Majumdar, “Transient ballistic and diffusive phonon heat transport in thin films,” *Journal of Applied Physics* **74**, 31–39 (1993).
- [5] J. M. Ziman, *Electrons and Phonons*, Oxford University Press (1960).
- [6] D. G. Cahill, W. K. Ford, K. E. Goodson, *et al.*, “Nanoscale thermal transport,” *Journal of Applied Physics* **93**, 793–818 (2003).

- [7] G. Chen, *Nanoscale Energy Transport and Conversion*, Oxford University Press (2005).
- [8] M. F. Modest, *Radiative Heat Transfer*, McGraw-Hill (1993).